

LAB 2

Solve following questions:

1. Write a program that takes a student's score as input and outputs the corresponding grade based on the following scale:

A: 90-100

B: 80-89

C: 70-79

D: 60-69

F: 0-59

PROGRAM:

```
package nDemo;

import java.util.Scanner;

public class examResult {

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.println("Enter Your Score : ");
        int examScore = sc.nextInt();
        sc.close();

        String grade;

        // if statement to check if the entered score is invalid
        if (examScore < 0 || examScore > 100) {
            System.out.println("Invalid score entered. Please enter a value
between 0 and 100.");
        } else {
            // Switch statement to evaluate marks in grades
            switch (examScore / 10) {
                case 10: // This will handle the case when score is 100
                case 9:
                    grade = "A";
                    break;
                case 8:
                    grade = "B";
                    break;
                case 7:
                    grade = "C";
                    break;
                case 6:
                    grade = "D";
                    break;
                default:
```

```

        grade = "F";
        break;
    }
    // Printing the grade
    System.out.println("The grade is: " + grade);
}
}
}

```

OUTPUT:

```

<terminated> examresult [Java Application] C:\Users\Asni cr
Enter Your Score :
88
The grade is: B

```

2. Write a program to check if a given year is a leap year. (A year is a leap year if it is divisible by 4 but not by 100, or it is divisible by 400.)

PROGRAM:

```

package nDemo;
import java.util.Scanner;

public class LeapYear {

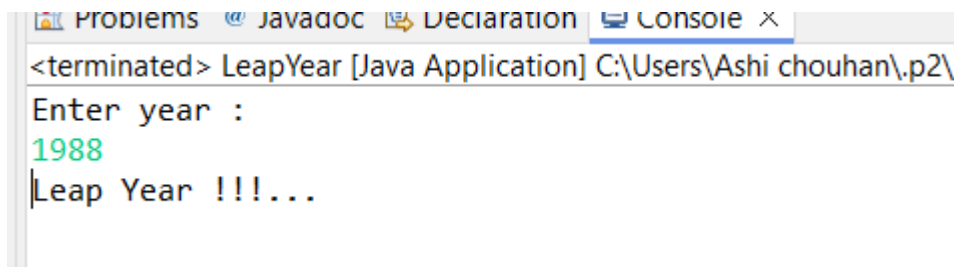
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);
        System.out.println("Enter year : ");
        int year = sc.nextInt();
        sc.close();

        // If statement to check if entered year is divisible by 100
        // If the year is divisible by 100 and 400 then it is not a
        // leap year
        if ( (year % 100 != 0) && (year % 400 != 0) ) {
            if ( year % 4 == 0 ) {
                System.out.println("Leap Year !!!...");
            }
        }
        else {
            System.out.println("Not a Leap Year !");
        }
    }
}

```

OUTPUT:



```
<terminated> LeapYear [Java Application] C:\Users\Ashi chouhan\.p2\
Enter year :
1988
Leap Year !!!...
```

3. Write a program that takes an integer as input and checks if it is positive, negative, or zero.

PROGRAM:

```
package nDemo;
import java.util.Scanner;

class IntegerType {

    public void typeIdentifier (int num ) {

        // If-else to check if a number is positive , Negative or Zero
        if ( num > 0 ) {
            System.out.println("Positive !!! ");
        }
        else if ( num < 0 ) {
            System.out.println("Negative !!!");
        }
        else {
            System.out.println("Zero !!!");
        }
    }
}

public class PositiveNegativeInteger {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        // Taking integer input from user
        System.out.println("Enter an Integer : ");
        int num = sc.nextInt();
        sc.close();

        // IntegerType class object
        IntegerType result = new IntegerType();
        result.typeIdentifier(num);
    }
}
```

OUTPUT:

```
<terminated> TestIntegerApplication.c
Enter an Integer :
-11
Negative !!!
|
```

4. Write a program that prints numbers from 1 to 10 using a loop.

PROGRAM:

```
package nDemo;

public class forLoopDemo {

    public static void main(String[] args) {

        int num = 1;

        // For loop to print 1-10 integer
        for(num = 1 ; num <= 10 ; num++ ) {
            System.out.println(num);
        }
    }
}
```

OUTPUT:

```
<terminated> forLoopDemo [Java]
1
2
3
4
5
6
7
8
9
10
```

5. Write a program that takes an integer N as input and calculates the sum of entered numbers.

PROGRAM:

```
package nDemo;
```

```

import java.util.Scanner;

public class sumOfEnteredNumber {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        // Taking user input for length of integer
        System.out.print("Enter the number of integers you want to sum: ");
        int N = sc.nextInt();
        int i = 0;
        int result = 0;

        // for loop to iterate the length if the integer
        for(i = 0 ; i < N ; i++ ) {
            if(i < N ) {
                System.out.println("Enter the " + (i + 1) + " : ");
                // Taking user input for integer
                int num = sc.nextInt();

                result += num;
            }
            else {
                System.out.println("Enter valid integer of same length as
entered before !");
            }

        }
        System.out.println("The sum is : " + result);
        sc.close();
    }
}

```

OUTPUT:

```

<terminated> sumOfEnteredNumber [Java Application] C:\Users\Ashi choi
Enter the number of integers you want to sum: 3
Enter the 1 :
1
Enter the 2 :
2
Enter the 3 :
3
The sum is : 6

```

6. Write a program that takes an integer as input and prints its multiplication table up to 10.

PROGRAM:

```

package nDemo;
import java.util.Scanner;

```

```

public class MultiplicationTable {

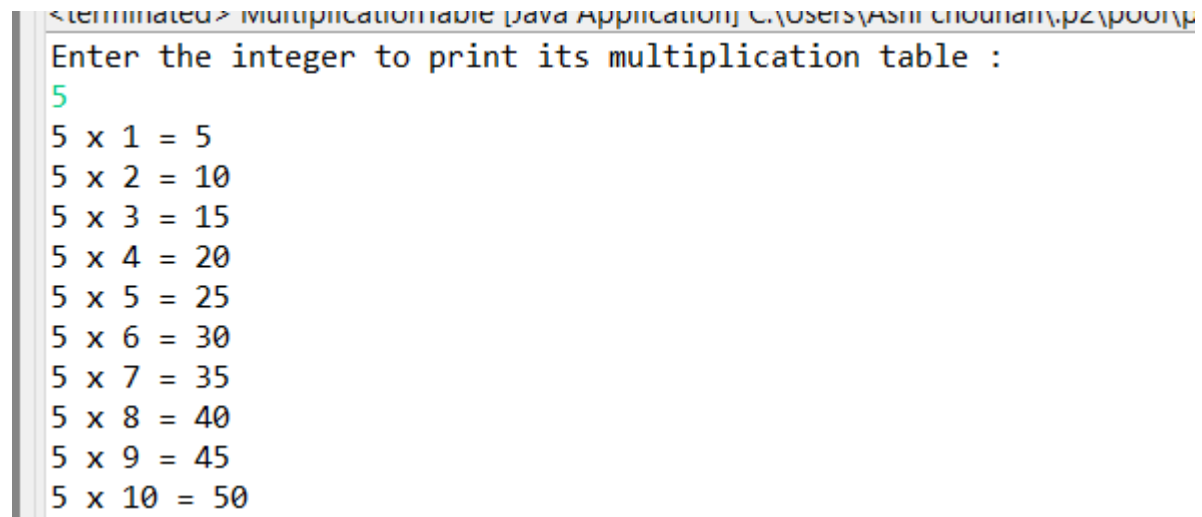
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);
        // Taking user input for integer
        System.out.println("Enter the integer to print its multiplication
table :");
        int num = sc.nextInt();

        // For loop to print multiplication table
        for(int i = 1 ; i <= 10 ; i++ ) {
            System.out.println(num + " x " + i + " = " + (num * i));
        }
    }
}

```

OUTPUT:



```

<terminated> MultiplicationTable [Java Application] C:\Users\Ashir Chowdhary\AppData\Local\Programs\Java\jdk-11.0.10\bin\java.exe
Enter the integer to print its multiplication table :
5
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
5 x 6 = 30
5 x 7 = 35
5 x 8 = 40
5 x 9 = 45
5 x 10 = 50

```

7. Write a program that takes a positive integer as input and prints its digits in reverse order.

PROGRAM:

```

package nDemo;
import java.util.Scanner;

public class ReverseDigits {

    public static int reverseDigits(int number) {
        int reversedNumber = 0;

        // While loop to check if the entered number is not 0.
        while (number != 0) {
            // Removing the unit place digit of number
            int digit = number % 10;
            // Storing unit place in reversedNumber
            reversedNumber = reversedNumber * 10 + digit;
        }
    }
}

```

```

        // Now the number is divided by 10 and stored in number variable
        number /= 10;
    }
    return reversedNumber;
}

public static void main(String[] args) {
    Scanner scanner = new Scanner(System.in);

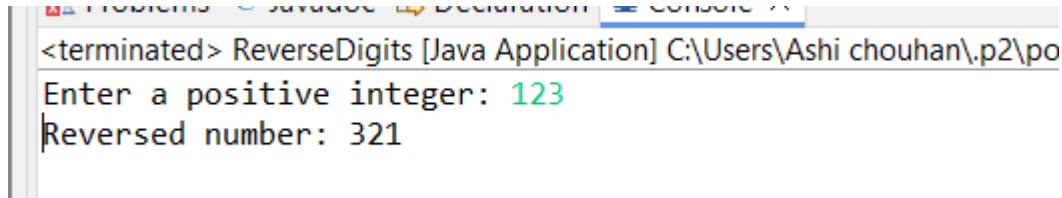
    // Taking user input
    System.out.print("Enter a positive integer: ");
    int userInput = scanner.nextInt();

    // Print warning statement if entered invalid integer
    if (userInput < 0) {
        System.out.println("Enter a positive integer !");
    } else {
        // Function call
        int reversedNumber = reverseDigits(userInput);
        System.out.println("Reversed number: " + reversedNumber);
    }

    scanner.close();
}
}

```

OUTPUT:



```

<terminated> ReverseDigits [Java Application] C:\Users\Ashi chouhan\.p2\po
Enter a positive integer: 123
Reversed number: 321

```

8. Create a class `Animal` with a method `makeSound()` that prints "Some generic animal sound". Create another class `Dog` that extends `Animal` and overrides the `makeSound()` method to print "Bark". Write a main method to demonstrate calling the `makeSound()` method on an `Animal` reference holding a `Dog` object.

PROGRAM:

```

package nDemo;

class Animal {
    // makeSound() function of Animal class
    void makeSound() {
        System.out.println("Some generic animal sound");
    }
}

class Dog extends Animal {
    // makeSound() function of Dog class
    void makeSound() {

```

```
        System.out.println("Bark");
    }
}

public class MainAnimal {

    public static void main(String[] args) {
        // Creating object of Dog class
        Dog d1 = new Dog();
        // Calling makeSound() function in Dog class
        d1.makeSound();
    }
}
```

OUTPUT:

